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Prediction of adverse drug reactions using demographic and non-clinical drug characteristics in FAERS data

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The presence of adverse drug reactions (ADRs) is an ongoing public health concern. While traditional methods to discover ADRs are very costly and limited, it is prudent to predict ADRs through non-invasive methods such as machine learning based on existing data. Although various studies exist regarding ADR prediction using non-clinical data, a process that leverages both demographic and non-clinical data for ADR prediction is missing. In addition, the importance of individual features in ADR prediction has yet to be fully explored. This study aims to develop an ADR prediction model based on demographic and non-clinical data, where we identify the highest contributing factors. We focus our efforts on 30 common and severe ADRs reported to the Food and Drug Administration (FDA) between 2012 and 2023. We have developed a random forest (RF) and deep learning (DL) machine learning model that ingests demographic data (e.g., *Age* and *Gender* of patients) and non-clinical data, which includes chemical, molecular, and biological drug characteristics. We successfully unified both demographic and non-clinical data sources within a complete dataset regarding ADR prediction. Model performances were assessed via the area under the receiver operating characteristic curve (AUC) and the mean average precision (MAP). We demonstrated that our parsimonious models, which include only the top 20 most important features comprising 5 demographic features and 15 non-clinical features (13 molecular and 2 biological), achieve ADR prediction performance comparable to a less practical, feature-rich model consisting of all 2,315 features. Specifically, our models achieved an AUC of 0.611 and 0.674 for RF and DL algorithms, respectively. We hope our research provides researchers and clinicians with valuable insights and facilitates future research designs by identifying top ADR predictors (including demographic information) and practical parsimonious models.

Keywords Adverse drug reactions, Deep learning, Random forest, Machine learning, Demographic features, Non-clinical features

An adverse drug reaction (ADR) is defined as "a response to a drug which is noxious and unintended and which occurs at doses normally used in man for prophylaxis, diagnosis, or therapy of diseases or for modification of physiological function"[1]. ADRs may result in detrimental consequences such as hospitalization, disability, congenital anomaly, or even be life-threatening[1]. The Institute of Medicine (IOM) reported that approximately 98,000 deaths occurred due to medical errors while nearly 7,000 deaths were a direct result of an ADR in the year 2000[2]. In another study, it was estimated that 6.3% of hospitalizations were due to ADRs from 2008 to 2011[3]. Typically, an ADR is discovered either in clinical trials or post-marketing surveillance. Although ADRs discovered in phase III of the clinical trials are considered the main adverse reactions (which are recorded in the product information), additional reactions, especially less common ones, may not be uncovered through the clinical trials due to issues such as limitations in patient selection, small sample sizes, and short time spans[4,5]. In contrast to the clinical trial phase, post-marketing surveillance, which heavily relies on voluntary reports provided by patients and health authorities, has a higher capability of identifying rare ADRs, particularly

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reactions that occur in specific populations or are brought about due to long-term drug exposure or adverse drug interactions[6]. Nevertheless, both approaches suffer from various drawbacks such as being very costly and time-consuming[7]. Therefore, there is an urgent need for other methods that can overcome these limitations. The availability of data sources provides an opportunity to develop predictive models in order to predict ADRs rather than just discovering them through clinical trials or post-marketing surveillance.

In recent years, several studies have been performed to develop analytical methods to predict ADRs[8–13], and the advancement of machine learning approaches have garnered much attention in applying them to discover hidden patterns between drugs and ADRs. Each drug includes unique non-clinical feature sets (e.g., chemical and biological) in which each set contains a multitude of properties. Moreover, each drug is associated with several ADRs. Therefore, ADR prediction models aim to utilize high-dimensional data in order to predict the multiple ADR outputs (multi-label classification problem).

Recently, several studies developed models that are capable of identifying reported ADRs from unstructured or semi-structured data sources. For example, Finkelstein, et al.[14] developed a method to automatically identify ADRs from journal articles, where they summarized and ranked the ADRs in a user-friendly interface. In another study, Pierce, et al.[5] developed a machine learning model in order to uncover drug-adverse pairs on social media. Similarly, Lee, et al.[15] developed a semi-supervised convolutional neural network to detect ADRs on Twitter. Munkhdalai, et al.[16] utilized three different machine learning models to extract ADR-related information from electronic health records narratives. Yahya, et al.[17] identified ADRs for anti-epileptic drugs from a medical forum (AskaPatient.com).

Moreover, several studies adopted different machine learning methods to predict ADRs that are often based on non-clinical features. Among all machine learning methods, deep learning (DL) is known to be highly effective for multi-label classification problems with high-dimensional datasets, where a few ADR studies[8,12] have exploited its use in overcoming limitations posed by traditional models. For instance, Wang, et al.[12] developed a DL model to predict ADRs based on the molecular descriptors, biological and biomedical properties of drugs. Similarly, Dey, et al.[8] developed a DL model that could predict ADRs and reveal the drug molecular structures related to those ADRs at the same time.

In addition, several studies have leveraged traditional machine learning methods to predict ADRs. Liu, et al.[11] developed five machine learning models (random forest (RF), support vector machine (SVM), naïve Bayes, K-nearest neighbor, and logistic regression) that predict ADRs using phenotypic, chemical fingerprint, and biological features. Furthermore, it has been shown that graph-structured data analytics methods achieve promising results in predicting ADRs[18,19].

Besides non-clinical features, demographic features play a critical role in predicting ADR occurrence, where it was shown that individuals from different demographics experienced dissimilar adverse reactions from the same drug[20]. Therefore, demographic features are crucial in predicting ADRs based on numerous studies[7,21]. For example, Lee, et al.[21] studied the role of gender in adverse drug reactions related to contrast media. In another study, Watson, et al.[22] discussed the impact of gender on ADRs, where they recommended studying its effect during the life-cycle of drug development and surveillance. Zekarias, et al.[23] investigated ADRs related to COVID-19 drugs and discovered that ADRs vary from males to females; therefore, they suggested further elucidation on the role of gender in ADRs. Hunt, et al.[24] analyzed age differences in drug-induced liver injury and recognized that acute liver failure reactions were more prominent in children (age 0–17) compared to other age groups. Alomar[25] determined that both age and gender are important factors regarding ADR occurrence, where very young or very old patients were found to be more susceptible to adverse reactions while women and men hold different biological characteristics that affect the occurrence of ADRs.

Although it has been shown that both demographic and non-clinical data play important roles in understanding ADRs, to the best of our knowledge, studies have yet to combine both demographic and non-clinical data for ADR prediction. Previous studies have incorporated either non-clinical data without demographic information of patients or clinical data alone to predict ADRs[7]. This paper aims to fill this gap by combining demographic data and non-clinical data to develop a practical model to predict ADRs. Our objective is also to facilitate future designs and insights among researchers and professionals in the area of healthcare by identifying the best predictors of ADRs and the impact of demographic features. We focus our study on predicting 30 ADRs derived from ADR severity lists produced by Wu, et al.[26]. Additionally, we analyze individual features from both demographic and non-clinical data as opposed to investigating a feature set (e.g., chemical, biological) in its entirety, performed in previous research. Analyzing individual features has two advantages: first, it provides information on the most impactful features of ADR prediction. Second, by identifying the most important features, we can reduce the complexity of the developed model, enhancing its practicality and improving its generalization capabilities.

In this study, we utilized both RF and DL classifiers to predict ADRs. The DL model is considered an advanced learning method while RF is a traditional ensemble learning method. One of the drawbacks of a DL model is its black-box nature which makes it hard to interpret the working mechanisms behind the model. Additionally, it prevents the extraction of important features and their contribution to the predictive model. In contrast, RF, despite being a less sophisticated method compared to a DL model, enables us to readily extract the importance of each individual feature and its contribution towards ADR prediction.

Results

In this study, we leveraged RF and DL multi-label classification models to predict 30 common and severe ADRs. We trained our models (80/20 splits) using all combinations of Chemical, Molecular, and Biological feature sets with and without the inclusion of a Demographic feature set. By integrating data from DrugBank and PubChem with the FDA Adverse Event Reporting System (FAERS) database from 2012 to 2023, we compiled a comprehensive and clean dataset consisting of 1,365,846 reported cases, 946 drugs, 881 chemical features,

17 molecular properties, 1,411 biological features, five *Age* subgroups, and *Gender*. Table 1 is structured to elucidate the inclusion effect of the demographic features (*Age* and *Gender*) on ADR prediction. We highlight in **bold** the performance metrics where the inclusion of demographic information increases AUC and MAP for ADR prediction. Several insights can be gleaned from the results in Table 1. First, regardless of feature set combination, 13 out of 15 models developed with DL have equal or higher performance compared to the RF method based on AUC. For example, the AUC value for ADR prediction using all feature sets is 0.681 and 0.700 for the RF and DL models, respectively. Additionally, 14 out of 15 developed models using DL have equal or larger MAP values compared to their RF variants. Second, the AUC value of a feature rich DL model with Chemical + Molecular + Biological feature sets (2,309 features) increases from 0.682 to 0.700 with the inclusion of 6 demographic features. Similarly, the AUC value of the DL model with Molecular + Biological feature sets (1,428 features) increases from 0.670 to 0.691 with the inclusion of the Demographic feature set. The AUC value of the DL model with the Chemical + Molecular feature sets (898 features) increases from 0.617 to 0.634 with the inclusion of the Demographic feature set. The AUC value of the DL model with only the Chemical feature set (881 features) increases from 0.624 to 0.637 with the inclusion of the Demographic feature set. The AUC value of the DL model with only the Molecular feature set (17 features) increases from 0.574 to 0.613 with the inclusion of the Demographic feature set. Third, by using only 6 demographic features, we achieve an AUC value of 0.555 using the DL model. These improvements in model performance, achieved by adding just 6 demographic features to extensive non-clinical feature sets, highlight the positive impact of *Age* and *Gender* on ADR prediction. In addition, the receiver operating characteristic (ROC) curves for all the listed RF and DL models in Table 1 are displayed in Figs. 1 and 2, respectively.

To quantify the impact of individual features on ADR prediction, we extracted the top 20 important features from the feature-rich RF model, which includes Chemical, Molecular, Biological, and Demographic feature sets. Additionally, we calculated the AUC value using RF and DL models for each individual feature for ADR prediction as shown in Table 2. Our analysis using RF feature selection revealed that patient demographics are among the most important features for predicting ADRs, with the top 5 out of 20 important features being demographic (*Age* subgroups and *Gender*). The remaining important features comprise 13 molecular and 2 biological features.

On the other hand, by comparing individual AUC values, some biological and molecular features appear to be more important than individual demographic features. This discrepancy can be attributed to the different ways in which RF feature importance and individual AUC values measure the contribution of features. RF feature importance captures the overall impact of features, including their interactions and non-linear relationships with other features[27], while individual AUC values measure the predictive power of each feature in isolation[28]. Consequently, while demographic features may not individually provide high AUC values, their synergistic contribution alongside other features is important in ADR prediction.

In order to gauge the difference in prediction performance between a model that incorporates only the topmost important features compared to one that uses all (2,315) available features, we trained RF and DL models using the top 20 features from Table 2. In addition, to assess the importance of demographic features in the parsimonious model (20 features) we trained another model, excluding demographic data. Our results in Table 3 show that RF and DL models utilizing the top 20 important features yield AUC values of 0.611 and 0.674, whereas, using feature-rich models (2,315 features) yield AUC values of 0.681 and 0.700, respectively. This result demonstrates that by combining the 5 demographic features and the reduced 15 non-clinical features (13 molecular + 2 Biological), a parsimonious DL model provides comparable performance in predicting ADRs compared to a feature-rich DL model. Moreover, the AUC value of the DL model with 13 molecular and 2

Feature sets	# of features	AUC		MAP	
		RF	DL	RF	DL
Chemical + Molecular + Biological + Demographic	2,315	**0.681**	**0.700**	0.430	**0.477**
Chemical + Molecular + Biological	2,309	0.671	0.682	0.430	0.460
Chemical + Biological + Demographic	2,298	**0.680**	**0.685**	0.427	**0.463**
Chemical + Biological	2,292	0.675	0.677	0.429	0.459
Molecular + Biological + Demographic	1,434	**0.691**	**0.691**	0.436	**0.458**
Molecular + Biological	1,428	0.687	0.670	0.437	0.448
Chemical + Molecular + Demographic	904	**0.623**	**0.634**	0.403	**0.417**
Chemical + Molecular	898	0.619	0.617	0.406	0.407
Biological + Demographic	1,417	**0.669**	**0.681**	**0.418**	**0.450**
Biological	1,411	0.667	0.670	0.417	0.447
Chemical + Demographic	887	**0.628**	**0.637**	0.399	**0.420**
Chemical	881	0.621	0.624	0.402	0.417
Molecular + Demographic	23	**0.569**	**0.613**	**0.360**	**0.366**
Molecular	17	0.554	0.574	0.359	0.351
Demographic	6	0.555	0.555	0.334	0.334

Table 1. RF and DL model performance utilizing all feature set combinations. Metrics where the inclusion of demographic information increases ADR prediction performnce are in bold.

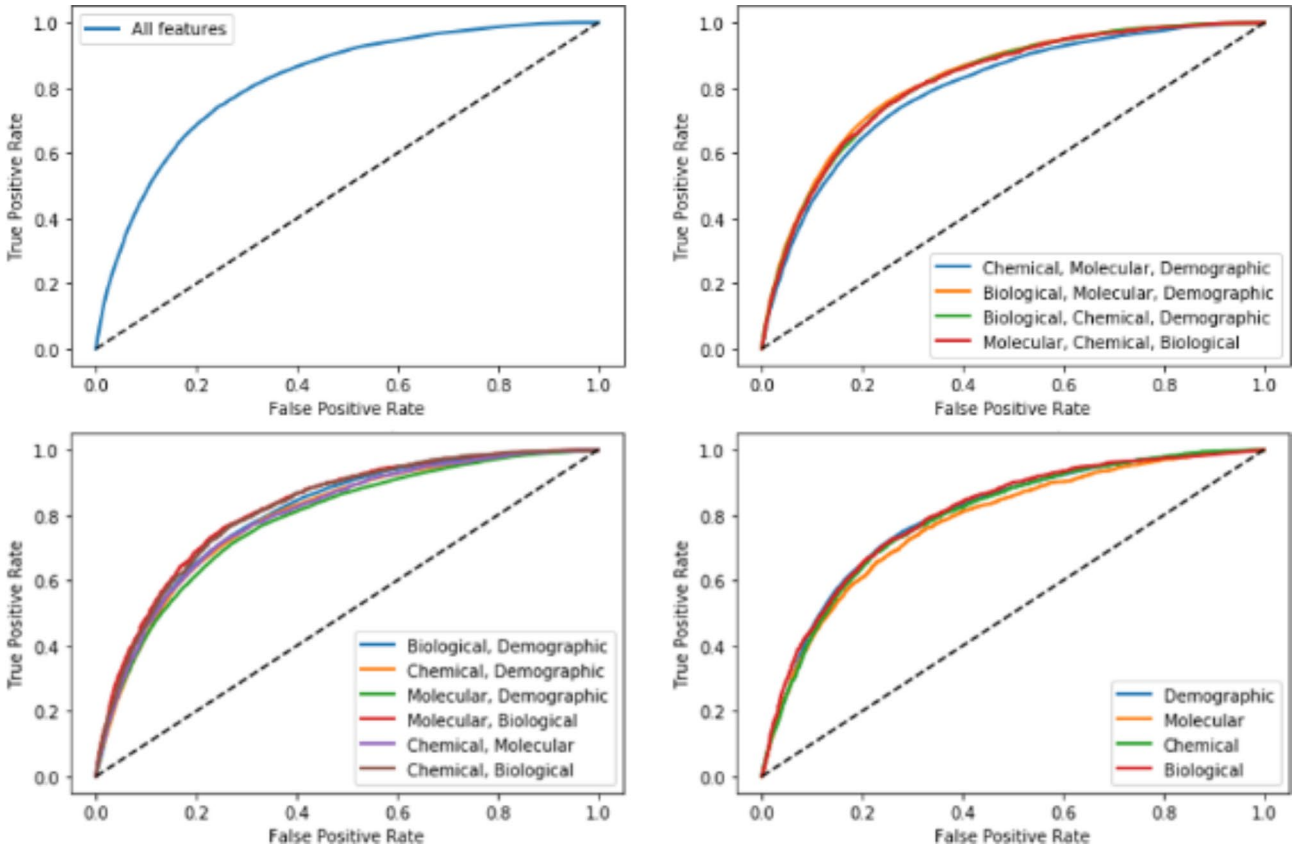


Fig. 1. ROC curves for RF models.

biological features increases from 0.652 to 0.674 with the inclusion of the demographic features. These findings again illustrate the importance of using demographic features when developing a practical and parsimonious model for predicting ADRs.

Materials and methods

Data

The datasets that are used in ADR studies can be broadly categorized into clinical and non-clinical data[7]. Non-clinical data provides information about the properties of drugs such as chemical (including substructure fingerprints and molecular descriptors), biological (including carriers, enzymes, targets, and transporters), and phenotypic characteristics (including therapeutic properties). Clinical data such as electronic health records and spontaneous reporting systems include ADRs reported after treatment and the demographic information of patients such as drug dosages, age, gender, and types of diseases[7]. In addition, some valuable ADR-related information might also be found in social media, medical forums, clinical narratives, or biomedical literature[29].

In this study, we collected our data from three main sources. The non-clinical features, which include biological and chemical properties of drugs, were extracted from DrugBank[30] and PubChem[31], respectively. Note that ADRs frequently occur due to interactions between drugs and specific proteins; therefore, uncovering these interactions is imperative in ADR-related studies[32–34]. The DrugBank dataset classifies drug proteins into four classes: drug targets, enzymes, carriers, and transporters[30]. We incorporated these four properties as biological features of the drugs in which each drug may possess multiple properties. A drug target is a protein in the body that is potentially associated with a specific disease and is targeted by a drug[35]. Drug enzymes affect drug metabolism and consequently induce ADRs[13]. Drug transporters admit a drug into and around the cell and play significant roles regarding the absorption, distribution and metabolism, excretion, and toxicity of drugs[36,37]. Drug carriers deliver drugs around the body in order to maximize drug efficacy, which can minimize adverse reactions[36,38]. The chemical properties are represented by substructure fingerprints and molecular descriptors. Substructure fingerprints are coded using a binary vector of 881 substructures where each vector component indicates the existence of a particular property[39,40]. Molecular descriptors are numerical values that describe chemical, physical, or topological properties of molecules such as molecular weight or atom count[39]. For the sake of convenience, we will refer to substructure fingerprints as "chemical features" throughout the remainder of this paper, although a molecular descriptor is also a type of chemical feature.

In order to incorporate demographic features and ADRs within our predictive models, we have also used the FAERS dataset collected by the FDA (openFDA, 2021). The FAERS data source is based on voluntary reports that were submitted by either healthcare professionals or patients (representatives). The FAERS dataset holds

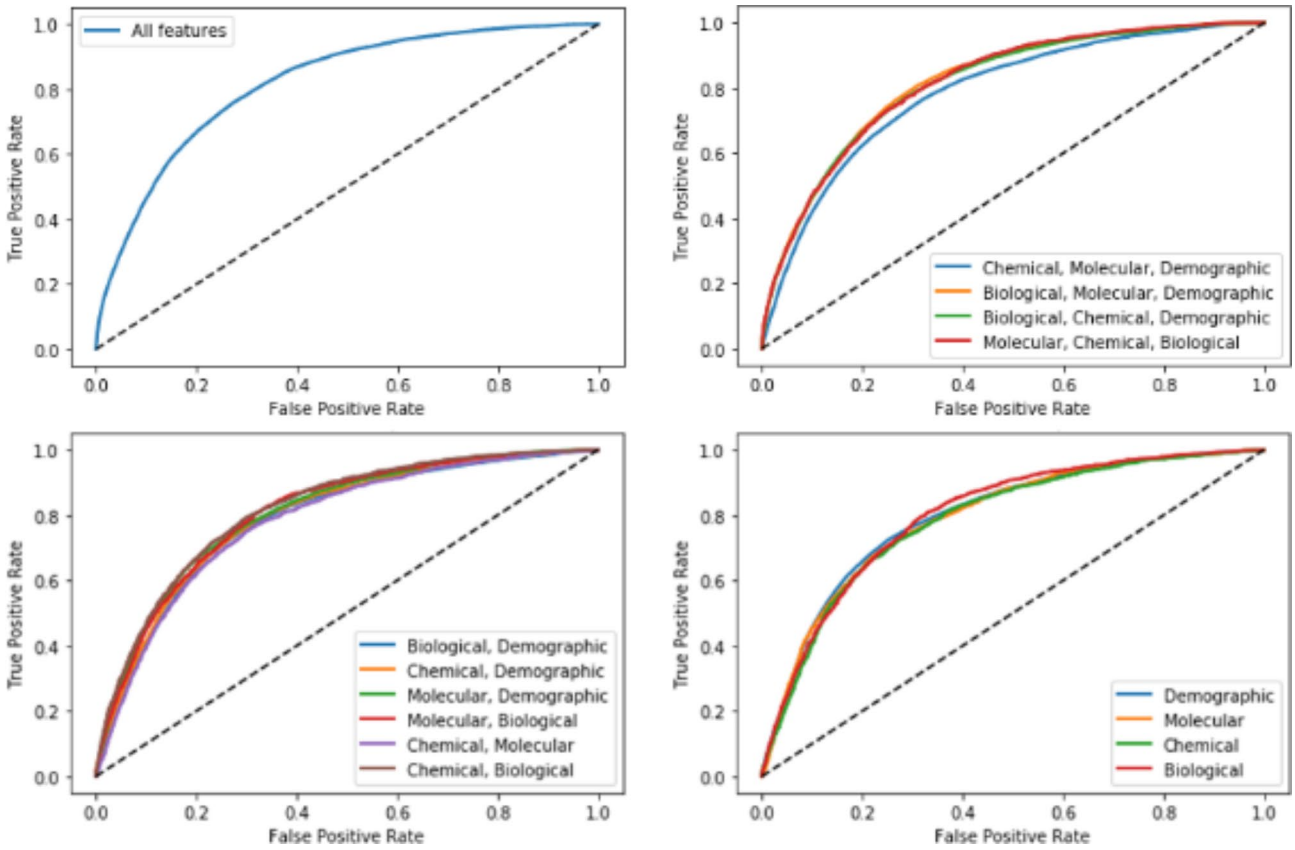


Fig. 2. ROC curves for DL models.

Rank	Feature set	Feature	Importance using RF	Individual AUC by RF	Individual AUC by DL
1	Demographic	Gender	0.063	0.513	0.509
2		Age: between 61 to 76 years	0.046	0.514	0.512
3		Age: older than 76 years	0.045	0.511	0.510
4		Age: between 40 to 61 years	0.045	0.516	0.516
5		Age: between 17 to 40 years	0.044	0.507	0.508
6	Molecular	XLogP3	0.011	0.516	0.539
7		Topological Polar Surface Area	0.011	0.518	0.550
8		Heavy Atom Count	0.011	0.499	0.529
9		Molecular Weight	0.011	0.507	0.529
10		Covalently Bonded Unit Count	0.011	0.511	0.527
11		Hydrogen Bond Acceptor Count	0.011	0.518	0.522
12		Monoisotopic Mass	0.011	0.509	0.530
13		Exact Mass	0.011	0.510	0.538
14		Rotatable Bond Count	0.011	0.493	0.496
15		Complexity	0.010	0.492	0.550
16		Hydrogen Bond Donor Count	0.010	0.490	0.514
17		Defined Bond Stereocenter Count	0.006	0.522	0.520
18		Undefined Bond Stereocenter Count	0.004	0.489	0.514
19	Biological	ABCB1	0.004	0.595	0.604
20		ALB	0.003	0.561	0.551

Table 2. Top 20 important features based on RF feature selection and their individual AUC values.

Features	AUC - RF model	AUC - DL model
All 2,315 features: Chemical + Molecular + Biological + Demographic	0.681	0.700
Top 20 important features: 5 Demographic + 13 Molecular + 2 Biological	0.611	0.674
Top features excluding Demographic set: 13 Molecular + 2 Biological	0.597	0.652

Table 3. RF and DL model performance utilizing all, top 20, and top non-clinical (exclusion of demographic set from top 20) features.

Feature	Age					Gender	
Group	< 17	17–40	41–60	61–76	> 76	Female	Male
Proportion	0.025	0.123	0.387	0.322	0.143	0.693	0.307

Table 4. *Age* and *gender* proportions within the clinical dataset.

information on patients and the ADR(s) they experienced. Specifically, FAERS contains 7 datasets which are mapped to one another: demographic, drug, patient outcome, reaction, drug therapy, indication and report source data[41]. The ADRs which are stored in the FAERS database are based on MedDRA's preferred terms. In contrast, the drug names included in the FAERS data are not standardized and may be referenced by their brand name, ingredient name, or even a typo[41]. We employ patient age and gender as demographic features in this study. As we discussed prior, both age and gender serve as the main risk factors regarding ADR occurrences.

Data pre-processing

In our study, we collected recent FAERS data spanning from 2012 to 2023. The FAERS database, initiated in 2012, succeeded the Legacy AERS, an earlier Adverse Event Reporting System. Given the importance of having accurate and standardized data, we emphasized rigorous data pre-processing[42]. Data pre-processing is one of the most important steps in the development of analytical prediction models since all of the remaining steps are built from this point on. Often, 80% of the time required for data analysis is spent on data pre-processing and cleaning[43].

Our data pre-processing procedure is comprised of five steps. First, the drug names listed in the FAERS database are not standardized. To address this, we applied the same drug name mapping methodology as Khaleel, et al.[44] which standardizes drug names using RxNorm. Second, to minimize model bias, we avoided imputation techniques and removed all patients with unknown demographic data (*Age* and *Gender*). Third, similar to previous studies[45,46], we categorized the *Age* feature into five groups: <17 years, 17–40 years, 41–60 years, 61–76 years, and >76 years, in order to measure the importance of each group. Fourth, we validated the drug-ADR pairs using the SIDER drug information database[47] and removed any pairs that could not be validated. Fifth, we performed data deduplication to ensure the integrity and accuracy of the dataset. Table 4 presents patient proportions in our dataset broken down based on *Age* and *Gender*.

Given the study's emphasis on predicting common and severe ADRs, we selected the top 30 ADRs from lists provided by Wu, et al.[26] who classify ADRs into three groups based on FDA-approved drug labeling: Boxed Warnings, Warnings and Precautions, and Adverse Reactions. Although there is some overlap among these groups, the severity of ADRs is highest in Boxed Warnings, followed by Warnings and Precautions, and then Adverse Reactions. Each list consists of the top 20 ADRs in each group, based on an analysis of 1,164 drugs[26]. We began our ADR selection process with the most severe Boxed Warnings group, selecting ADRs that corresponded with at least 400 patients of varying demographics to ensure a sufficient sample size[48] and to enhance the trainability of our predictive models. We then continued the selection process with the Warnings and Precautions group, followed by the Adverse Reactions group, ultimately selecting the top 30 ADRs with adequate sample sizes.

After the data preprocessing stage, the dataset contains 1,365,846 reported cases and 946 drugs. The feature sets include 881 chemical features, 17 molecular properties, 1,411 biological features, five *Age* subgroups, and *Gender*. Except for molecular properties, all feature sets are represented as binary vectors, where 1 indicates the presence of that feature in the drug. Additionally, Table 5 shows the 30 selected ADRs and their respective incidence proportions. Detailed visualizations of these ADR incidences, categorized by *Age* subgroups and *Gender*, are available in the Supplementary Information.

Machine learning models and evaluation metrics

In this study, we did not employ a conventional random train-test split. Instead, we partitioned the dataset at the molecular class level to prevent data leakage, ensuring that similar compounds with minor differences do not appear in both the train and test sets. To achieve this molecular class split, we applied K-means clustering on the molecular dataset, creating 25 distinct clusters. The number of clusters was determined based on two criteria. First, we utilized the within-cluster sum of squares (WCSS) method, which measures the sum of squared distances between each point and the centroid of its cluster. A lower WCSS indicates better clustering. We selected the number of clusters to ensure it is beyond the "elbow" point in the WCSS plot, where the rate of decrease in WCSS slows sharply. Second, we generated enough clusters to ensure that approximately 20% of

ADR	Proportion	Boxed Warnings	Warnings and Precautions	Adverse Reactions
Depression	0.025	X	X	
Infection	0.022	X	X	X
Myocardial infarction	0.011	X		
Haemorrhage	0.007	X	X	
Cardiac failure	0.007	X	X	
Cerebrovascular accident	0.007	X		
Completed suicide	0.007	X		
Coma	0.004	X		
Psychotic disorder	0.003	X		
Neoplasm malignant	0.001	X		
Hepatotoxicity	0.001	X		
Respiratory depression	0.001	X		
Nausea	0.137		X	X
Diarrhoea	0.119		X	X
Pain	0.090		X	X
Vomiting	0.084		X	X
Hypotension	0.035		X	X
Hypertension	0.031		X	X
Hypersensitivity	0.028		X	X
Renal failure	0.028		X	
Discomfort	0.011		X	
Renal impairment	0.009		X	
Diabetes mellitus	0.007		X	
Liver disorder	0.002		X	
Headache	0.104			X
Dizziness	0.083			X
Rash	0.070			X
Pruritus	0.049			X
Oedema	0.010			X
Nervousness	0.006			X

Table 5. Top 30 ADRs from FAERS (2012–2023) based on selection criteria.

the dataset would be allocated to the test set. After clustering, we performed a random split at the cluster level, resulting in a roughly 80/20 split.

We applied RF and DL models to ADR prediction, where both models are found to be highly applicable to multi-label classification problems with high-dimensional datasets (Wang et al., 2019). Random forest is a type of ensemble learning which utilizes multiple decision trees to make a prediction. In the RF classifier, we set the number of trees to 300 for ADR prediction since it performed well in our preliminary analysis. For the DL model, we implemented a 4-layer architecture where an input layer and output layer are connected via two hidden layers, which behave as higher-level representations of the input. The output layer consists of 30 nodes corresponding to 30 ADRs. Additionally, sigmoid activation functions are used in conjunction with cross-entropy serving as the loss function. We ran a total of 15 models ($2^4 - 1 = 15$ given 2 methods and 4 feature sets: Chemical, Molecular, Biological, and Demographic) using a combination of feature sets in each RF and DL model. In line with prior ADR research such as Wang, et al.[12] and Hu, et al.[49], we adopt the area under the receiver operating characteristic curve (AUC) and mean average precision (MAP) as performance metrics.

Discussion

Overview

The open-access option to medical data sources provides an opportunity to develop analytical models for ADR prediction using different types of drug properties. However, drug properties themselves are not the only factors that determine ADRs. A specific drug can produce varying ADRs for different patients, exemplifying the importance of patient demographics along with drug properties for reliably predicting ADRs. Additionally, evaluating feature importance is imperative to understanding which factors govern ADRs. To the best of our knowledge, we are the first study to combine patient demographic and drug property data regarding ADR classification. In this study, we made use of demographic data, spanning 12 years in length, from the FAERS data source together with biological properties, chemical properties, and molecular descriptors (non-clinical features) in order to predict the top 30 ADRs. We implemented RF and DL models for all combinations of feature sets to assess their effect on ADR prediction. Furthermore, we extracted the top 20 important features from the

main four feature sets by means of RF feature selection and developed RF and DL models based on these top 20 most important features (5 Demographic + 13 Molecular + 2 Biological) for predicting ADRs.

Principal findings

Based on the results from ADR classification models and feature importance analysis, this study makes two main contributions. First, we demonstrated that including just 6 demographic features enhances the AUC values for ADR prediction across various DL models. DL models with Chemical + Molecular + Biological, Molecular + Biological, and Molecular feature sets gained an increase of 1.8–3.9% in AUC values after incorporating demographic data, highlighting the positive impact of *Age* and *Gender* on model performance. Specifically, the AUC increased from 0.682 to 0.700 for the DL model with combined Chemical + Molecular + Biological feature sets, from 0.670 to 0.691 for the DL model with Molecular + Biological feature sets, and from 0.574 to 0.613 for the DL model with only the Molecular feature set. Furthermore, using only 6 demographic features, we achieved an AUC value of 0.555 with the DL model. These improvements in model performance, achieved by adding only 6 demographic features to extensive non-clinical feature sets, underscore the positive impact of including *Age* and *Gender* in ADR prediction.

Second, we showed that a parsimonious DL model, combining 5 demographic features with a reduced set of 15 non-clinical features (13 molecular and 2 biological), performs on par with a feature-rich (2,315 features) model in predicting ADRs. Based on RF feature selection results, patient demographics (*Age* subgroups and *Gender*) are among the top 20 important features after reducing all 2,315 features. This underscores the efficiency and effectiveness of a streamlined approach, proving that a model incorporating key demographic and selected non-clinical features can achieve robust ADR prediction without compromising accuracy. Importantly, our findings regarding the critical role of demographic factors such as *Age* and *Gender* align with existing literature on ADR studies.

Clinical studies have discussed that women are 1.5 to 1.7 times more at risk of ADRs than males[50,51]. Similarly, Rodenburg, et al.[52] found that female hospitalization due to ADRs is more prevalent than in men. Note that our dataset contains more reported ADRs among females than males (Table 4). There is compelling evidence that these differences are rooted in hormonal differences between women and men[53]. In addition, several prior studies have explored the role of age in ADR occurrences. For example, a study conducted by Yu, et al.[54] discovered that diarrhea and antibacterial symptoms happen more often in children, while dizziness is common in elderly people. In another study, de Araújo Lobo, et al.[55], showed that 15 to 50 year old adults have a higher risk of ADRs than any other age group.

In addition to features within the demographic dataset, various molecular features behave as strong predictors. The most important molecular feature is *XLogP3*, a parameter Log(P) that assesses lipophilicity, which refers to the compound's ability to attract fats, oils, and lipids[56]. Lipophilicity plays a significant role in describing drug properties such as absorption, distribution, metabolism, excretion, and toxicity (ADMET)[57]. In fact, predicting ADMET is one of the promising approaches in order to determine the efficacy and adverse reactions of drugs[30]. The role of Log(P) in drug efficacy and toxicity has been widely investigated in literature[58]. A high value for Log(P) pertains to an increased likelihood of a drug's toxic event, while a decrease in solubility and metabolism[58]. In another study, Chen, et al.[59] showed that taking a drug with high lipophilicity at a high dose can increase the risk of drug-induced liver injury. Similarly, Valko, et al.[60] discussed that higher lipophilicity reduces drug efficiency, and therefore, the need for a higher dose can increase the risk of ADRs.

The next important molecular feature is the *Topological Polar Surface Area*, a molecular polarity descriptor[61]. The molecular polar surface area of a drug helps characterize its transport properties such as intestinal absorption and blood – brain barrier crossing[62,63]. Hughes, et al.[64] discovered a negative relationship between polar surface area and the adverse effects of the drug.

Additional relevant molecular features are *Heavy Atom Count*, *Molecular Weight*, and *Complexity* as seen in Table 2. According to Chagas, et al.[65] both molecular complexity and weight features influence ADR development, where molecular weight is defined as the sum of all atomic weights. In terms of molecular complexity, two molecules with a similar weight may hold varying compositions in which one possesses higher numbers of atoms with lighter weight while the other displays a simple structure with few heavy atoms[65]. Nevertheless, many studies use molecular weight or heavy atom count as a substitution for complexity[66]. The correlation between molecular weight and the risk of an ADR has yet to be fully explored[58]. However, it was shown that increases in molecular weight and Log(P) can suppress all ADMET characteristics[67]. As for complexity, Sharma, et al.[68] have pointed out that ADR occurrences frequently coincide with more complex drug molecules. In addition, Lovering, et al.[69] discussed that molecules with more than three aromatic rings show higher risk of toxicity.

The most important biological feature is the *ABCB1* transporter, which is one of the ABC transporter members. ABC transporters are the main family in drug transport proteins that play key roles in drug distribution within the cells[70,71]. These transporters are distributed in various tissues, especially tissues that are responsible for drug absorption, metabolism, and excretion. Any changes in the function of these transporters increase the risk of ADR[70].

The second most important biological feature is *ALB* (albumin), a protein component of blood plasma produced by the liver. Albumin serves as a carrier for naturally occurring substances such as fatty acids and bilirubin, as well as external compounds including medications and nutrients[72]. Many drugs such as phenytoin and ibuprofen, bind to proteins like albumin to circulate within the bloodstream, exerting their effects through their free or unbound part. Lower levels of albumin result in a higher proportion of free drug, which can increase the risk of ADRs. Aging is also associated with a reduction in liver size and blood flow by up to 40%, impacting the metabolism of drugs that are extensively processed by the liver, such as diltiazem, opiates, and warfarin. Consequently, these drugs are less metabolized during the first pass through the liver, leading to elevated blood levels and more pronounced effects in older patients[73].

Limitations

Due to the complexity of ADR data, various challenges exist. First, to ensure sufficient sample sizes and enhance the trainability of our predictive models, we focused our analysis on common and severe ADRs associated with at least 400 patients of varying demographics in our dataset. Second, employing the methodology of Khaleel, et al.[44] to curate data necessitated omitting certain ill-defined drug names to maintain data quality and integrity. Third, while we analyzed a large dataset of over 1,365,846 cases, 946 drugs, and 30 ADRs, some prediction biases may remain due to data diversity, missing values, human factors in reporting, and unreported concomitant medications. For future work, the number of patients, drugs, and ADRs analyzed can be expanded. Furthermore, exploring the effects of demographic factors such as BMI and race, medical history, and social contexts like smoking and drinking habits on ADR prediction could provide valuable insights.

Conclusions

In this study, we successfully developed RF and DL models for ADR prediction by leveraging both demographic and non-clinical data, focusing on 30 common and severe ADRs reported to the FDA. Our analysis makes two main contributions: first, we demonstrated that including just 6 demographic features enhances the AUC values for ADR prediction across various DL models, highlighting the positive impact of *Age* and *Gender* on model performance. Specifically, models with Chemical + Molecular + Biological, Molecular + Biological, and Molecular feature sets gained an increase of 1.8–3.9% in AUC values after incorporating demographic data. Second, we showed that a parsimonious DL model combining 5 demographic features with a reduced set of 15 non-clinical features performed on par with a feature-rich model (2,315 features), underscoring the efficiency and effectiveness of a streamlined approach. These findings not only align with existing literature but also fill a critical gap by demonstrating the importance of combining demographic and non-clinical data to develop parsimonious ADR prediction models, paving the way for future research designs and insights among healthcare researchers and professionals.

Data availability

The raw datasets used in this study are accessible to the public on the following platforms: DrugBank (https://go.drugbank.com), PubChem (https://pubchem.ncbi.nlm.nih.gov), and the National Bureau of Economic Research (https://www.nber.org/research/data/fda-adverse-event-reporting-system).

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A.F.: Conceptualization, Methodology, Formal Analysis, Visualization, Writing - Original Draft. Z.S-M.: Conceptualization, Methodology, Writing - Original Draft. B.R.: Formal Analysis. J.J.H.: Conceptualization, Writing - Original Draft, Writing - Review & Editing. B.F.: Data Collection, Data Curation, Visualization.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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